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The Hong Kong University of Science and Technology, Hong Kong

RESEARCH EXPERIENCE

• Independent Research

March 2026 - Ongoing

Hong Kong University of Science and Technology

Hong Kong

- IR project name: CVPR/ICCV, supervised by Dr. Chi-Keung Tang

* FLAIR (Controllable Surgical Video Generation)

- Existing Problem: Reliance of current surgical T2V models on auxiliary visual conditions to generate surgical videos with stable and action-consistent motion.
- Our solution: We introduce FLAIR, the first reference-free framework for surgical video generation, which learns flow-derived latent action priors from real videos, predicts them from text prompts, and injects them into a frozen diffusion transformer to achieve stable and action-consistent motion, under reference-free conditions. Superior experimental results are achieved on key backbones including Wan-2.1, CogVideoX-2B, and HunyuanVideo-1.5.
- Our contribution also includes 1. SurgActionClip-30K, the first large-scale action-centric surgical video dataset spanning 11 source datasets with structured action-instrument-target captions, and 2. SurgMetrics, the first surgical-domain evaluation suite with six artifact-specific axes achieving strong alignment with human judgment.
- Role: First Co-first Author, paper submitted to **AAAI2027**

• Undergraduate Research Opportunity Program (UROP)

June 2025 - Ongoing

Hong Kong University of Science and Technology

Hong Kong

- UROP project name: Generative AI, supervised by Dr. Chen Qifeng, and Dr. Runtao Liu

* LiveStockArena (Large Language Model & Benchmark)

- Existing Problem: Nowadays, there is a growing population of products making use of AI Large Language Models to perform financial stock prediction tasks. Yet, there is a lack of a comprehensive benchmark for evaluating the stock prediction capability across different AI models under fair conditions in the community.
- Our solution: We have thus created a First dynamic benchmark for real-time AI model stock prediction by making use of the unpredictability of time. The benchmark is naturally resistant to cheating and evaluates models' genuine forecasting capabilities against live financial market.
- Research result: Collected 5000+ results across 5 months, serving as valuable resources for evaluating AI models' real-time stock prediction capabilities to the public. On top of that, we propose the first live benchmark system and build a leaderboard website, GitHub: <https://punktheory.github.io/leaderboard/>. Ongoing work focuses on extending the benchmark to AI agents beyond LLMs, benchmarking the harness to plan and execute trades.
- Role: Co-first Author, paper submitted to **EMNLP2027**

* SafetyDPO (Generative Text-to-Image AI Safety)

- Existing Problem: Current open-source T2I models are inherently capable of generating harmful content (e.g. violent and sexually explicit material), yet existing safety methods fail to suppress such outputs without compromising generative capability.
- Our solution: We transplanted the Direct Preference Optimization (DPO) algorithm from the LLM domain to T2I, training a LoRA expert under a custom-designed architecture to finetune a Stable Diffusion T2I model.
- Research result: The results show significant enhancement in safety for Stable Diffusion 1.5 models with an appropriate rate 93% (baseline 70%) while retaining almost full generative capability of the original models. Github: <https://github.com/Visualalignment/SafetyDPO>. Continuous work has been done to implement the architecture on Stable Diffusion 3.5 and yielding promising results as well.
- Role: Worked as an undergraduate research assistant (not author); paper accepted by ICCV 2025

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [S.1] **Tsz-Yui Qin**, Siyu Zhou, Chi-Keung Tang, Yuxiang Nie, Shu Yang. “**Beyond Masks and Trajectories: Flow-Guided Latent Action Injection for Stable Surgical Video Generation**” Submitted to AAAI 2027. (Under Review)
- [S.2] Runtao Liu, **Tsz-Yui Qin**, I Chieh Chen, Bo Yang, Qingping Yang, Jiahao Zhan, Qifeng Chen. “**LiveStockArena: Can Large Language Models Forecast Financial Markets?**” Submitted to EMNLP 2027. (Under Review)

EDUCATION BACKGROUND

- **The Hong Kong University of Science and Technology (HKUST)** Sep 2023 - Sep 2027(expected graduation time)
Bachelor of Engineering major in Computer Science Hong Kong
 - 103/120 credits completed
 - GPA: 3.158/4.3
 - Taken course: Computer Vision(A+), FYP(A+), Machine Learning, Reinforcement Learning
 - Taken course: Design of Algorithms, Object-Oriented Programming, Software Engineering
 - Research Interests: Computer Vision and Machine Learning, including Video Diffusion Model(VDM), Multimodal LLMs (MLLMs), Large Language Models (LLMs) and Agents

INTERNSHIP EXPERIENCE

- **Ant Group Co., Ltd.** June 2026 - August 2026
Position: Machine Learning Engineer (Research and Deployment of Agents) 
 - Research and development of industrial-grade **Agent** deployed in production, applied to Financial Risk Control workflows to automate risk assessment, transaction monitoring, and anomaly detection at scale.
 - Conceived and implemented a comprehensive **Agent benchmark evaluation framework**, driving systematic quantitative assessment and iterative improvement of agent to enhance agent quality
 - Participate in an agent-driven self-iteration system project, designing and implementing an autonomous upgrading mechanism that enables AI agents to self-optimize without manual intervention

ADDITIONAL INFORMATION

Languages: Cantonese and Mandarin (Native), English (Proficient)
Technical Skills: Python, PyTorch, HuggingFace, OpenCV, Github, Linux, Claude Code, Zotero
Leadership Experience: School Engineering Student Ambassador